P-C-1891
September 27, 1976
SUPERSEDING
Fed. Spec. P-C-450b
November 8, 1960

FEDERAL SPECIFICATION

CLEANING COMPOUND, PORCELAIN AND CERAMIC TILE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers three types of cleaning compounds for use in cleaning porcelain, ceramic tile, and related surfaces.

1.2 Classification. The cleaning compounds shall be of the following types, as specified.

Type I - Aerosol foam cleaner.
Type II - Cream cleanser.
Type III - Acid solution cleaner.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications.

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-96 - Cans, Metal, 28 Gage And Lighter.

Federal Standard.

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 7930

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:19Z
Check the source to verify that this is the current version before use.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

16 CFR 1500 - Hazardous Substances & Articles; Administration & Enforcement Regulations.
40 CFR 162 - Regulations for the Enforcement of the Federal Insecticide, Fungicide, and Rodenticide Act.

(The Code of Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society for Testing and Materials (ASTM) Standards:

D 1824 - Apparent Viscosity of Plastisols and Organosols at Low Shear Rates by Brookfield Viscometer.
E 70 - pH of Aqueous Solutions with the Glass Electrode.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Type I. The type I cleaner shall be an aerosol delivery type product consisting of a stable single phase liquid detergent solution or emulsion.

3.1.2 Type II. The type II cleaner shall be a free flowing liquid consisting of detergents, builders, and suspended abrasive particles. The abrasive particles shall be of a hardness not greater than that of silica. The product shall be stable without separation, settling, or caking.

3.1.3 Type III. The type III product shall be a liquid solution consisting of hydrochloric acid, wetting agents, and inhibitors. The product shall be clear and shall have no evidence of sediment.

3.2 pH. The pH of types I and II shall be in the range 7.0 to 10.0. Testing shall be in accordance with 4.4.1.3.

3.3 Cleaning efficiency.

3.3.1 Type I. When tested as specified in 4.4.1.4, the product shall demonstrate a cleaning efficiency of 50 percent or greater.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:19Z
Check the source to verify that this is the currentversion before use.

3.3.2 Type II. When tested as specified in 4.4.1.4, the product shall demonstrate a cleaning efficiency of 75 percent or greater.

3.4 Deleterious effects.

3.4.1 Steel.

3.4.1.1 Type I. When tested as specified in 4.4.1.5, the product shall not cause discoloration or etching of stainless steel.

3.4.1.2 Type II. When tested as specified in 4.4.1.6, the product shall not cause discoloration or etching of stainless steel.

3.4.1.3 Type III. When tested as specified in 4.4.1.7, the product shall not cause a weight decrease in the steel sample in excess of 0.15 percent.

3.4.2 Porcelain enamel. When tested as specified in 4.4.1.8, the type II product shall show no evidence of having etched the porcelain enamel.

3.5 Foam stability, type I. When tested as specified in 4.4.1.9, the foam height shall be 7 ml or greater, and shall decrease by 2 ml or less after 5 minutes.

3.6 Particle size, type II. When tested as specified in 4.4.1.10, a minimum of 0.5 grams shall be retained on a No. 325 sieve and a maximum of 0.25 grams shall be retained on a No. 80 sieve out of a 10 gram sample of cleaner.

3.7 Rinsability, type II. When tested as specified in 4.4.1.11, the product shall be completely removed from the flask and there shall be no evidence of a film deposit after drying.

3.8 Viscosity, type II. When tested as specified in 4.4.1.12, the viscosity of the product shall be a minimum of 2500 centipoises and a maximum of 4500 centipoises.

3.9 Non-volatile matter content, type II. The non-volatile matter content of the product shall be not less than 36 percent when tested as specified in 4.4.1.13.

3.10 Hydrogen chloride content, type III. The hydrogen chloride content of the product shall be not less than 5 percent and not more than 10 percent when determined as specified in 4.4.1.14.

3.11 Registration. If the product label states that the product has antimicrobial ingredients or effect, the product shall be registered with the Environmental Protection Agency in accordance with 40 CFR 162, Regulations for the Enforcement of the Federal Insecticide, Fungicide, and Rodenticide Act. When required, (see 6.2) the product shall be registered with the Environmental Protection Agency as a sanitizer or disinfectant, as applicable (see 6.2).

3.12 Labeling. When the product is not required to be registered with the Environmental Protection Agency (see 3.11), the labeling shall be in accordance with the applicable subparagraph below and with 16 CFR 1500, Hazardous Substances and Articles; Administration and Enforcement Regulations.

3.12.1 Type I label. The type I product label shall include adequate instructions for use, state the types of surfaces on which the product is intended to be used, and include the appropriate antidote procedures and caution statements.

3.12.2 Type II label. The type II product label shall include adequate directions for use, explaining the application, scrubbing and removal procedures, and adequate caution statements.

3.12.3 Type III label. The type III product label shall include directions for use and state the types of surfaces and materials which would be damaged by the product. The label shall also include adequate caution statements and antidote procedures.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:19Z
Check the source to verify that this is the current version before use.

contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection sampling. Sampling for the inspection procedures set forth in this specification shall be performed in accordance with MIL-STD-105. The lots to be inspected shall consist of no more than one classification (see 1.2) and one package size (see 5.1).

4.3 Examination.

4.3.1 Examination of preparation for delivery.

4.3.1.1 Sampling procedure. The sample unit shall be defined as one shipping container fully marked and prepared for delivery. The samples shall be selected in accordance with inspection level S-2 and the AQL shall be 4.0 percent defective.

4.3.1.2 Examination procedure. An examination shall be made to determine the compliance with the packing and marking requirements of section 5. The sample units shall be examined for the defects described below and scored accordingly.

Examination of preparation for delivery

Examine	Defects
Packing	Not as specified; odd size or shape of container.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of packing containers, or of component materials.
Marking	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.

4.3.2 Examination of the filled unit container.

4.3.2.1 Sampling procedure. The sample unit shall be defined as one filled container. The sample units shall be selected in accordance with inspection level S-3 and the AQL shall be 4.0 percent defective.

4.3.2.2 Examination procedure. An examination shall be made to determine the compliance of the contents and packaging with the requirements of 3.1 and section 5. The sample units shall be examined for the defects described below and scored accordingly.

Examination of the filled unit container

Examine	Defects
Contents	
Type II	Not free flowing; presence of lumps; evidence of separation.
Type III	Not transparent, presence of sediment.
Packaging	Not as specified.
Marking	Not as specified (see 3.12 and 5.3).

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T11:19Z
Check the source to verify that this is the currentversion before use.

4.4 Testing.

4.4.1 Cleaner testing.

4.4.1.1 Sampling plan. The sample unit shall be defined as one unit container. For type I, the average of the determinations of two sample units shall be used in determining compliance. For types II and III, a composite shall be made by mixing equal portions of each of the sample units taken from the lot in accordance with inspection level S-1. For type II, the composite shall be 2 pounds and for type III, the composite shall be 1 quart. The lot shall be unacceptable if the composite sample, or the type I average, fails to pass any of the applicable tests described in 4.4.1.2.

4.4.1.2 Testing procedure. The samples shall be tested in accordance with the instructions in table II.

TABLE II. Instructions for testing the porcelain cleaner

Characteristic	Applicable to type	Spec. Reference Requirement	Test Method	Results reported as Pass or Fail	Numerically to nearest	
pH	I, II	3.2	4.4.1.3		0.1	
Cleaning efficiency	I	3.3.1	4.4.1.4		5	percent
	II	3.3.2	4.4.1.4		5	percent
Deleterious effects, steel	I	3.4.1.1	4.4.1.5	X		
	II	3.4.1.2	4.4.1.6	X		
	III	3.4.1.3	4.4.1.7		0.01	percent
Deleterious effects, porcelain enamel	II	3.4.2	4.4.1.8	X		
Foam stability	I	3.5	4.4.1.9		1	ml
Particle size	II	3.6	4.4.1.10		0.5	percent
Rinsability	II	3.7	4.4.1.11	X		
Viscosity	II	3.8	4.4.1.12		1	centipoise
Non volatile matter	II	3.9	4.4.1.13		2	percent
Hydrogen chloride content	III	3.10	4.4.1.14		1	percent

4.4.1.3 pH. For type I, 10 grams of the liquid product from a depressurized container shall be mixed with 990 ml of distilled water. For type II, a 1 percent solution shall be made. The pH of the solutions prepared as specified above shall be determined as specified in ASTM E 70.

4.4.1.4 Cleaning efficiency, type I. Mix equal parts of vegetable shortening, lard, and cooking oil, using heat to homogenize the mixture. Weigh 0.2 g of the above soil mixture and spread evenly on a white porcelain test panel. Heat the panel in a forced air oven at 204° ± 1°C for 45 minutes and cool to room temperature. Spray enough of the cleaner on the panel to wet it thoroughly and scrub lightly with a damp sponge until the rate of soil removal has stopped (about 1 minute). Estimate by visual inspection the percentage of soil removed. The porcelain can be obtained as # 7812-S34 from the Arthur H. Thomas Co., P.O. Box 779, Philadelphia, PA 19105.

4.4.1.5 Deleterious effects on stainless steel, type I. Spray about 20 ml of the product in a 50 ml beaker and allow to liquefy. Immerse a stainless steel coupon in the product for 4 hours. Remove the coupon, dry, and examine for discoloration or etching using an untreated coupon for comparison. Disregard corrosion occurring at the air sample interface. The stainless steel can be obtained from Torrington Brass and Steel Co., Inc., P.O. Box 986, Torrington, CT 06790; steel part number: T-302, 1/2 hard, 0.62" x 1.375".

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:19Z
Check the source to verify that this is the current version before use.

4.4.1.6 Deleterious effects on stainless steel, type II. Place 20 ml of the sample in a 50 ml beaker. Immerse one end of a stainless steel coupon in the product for 4 hours. Examine for evidence of etching or discoloration using an untreated sample of stainless steel for comparison. Disregard corrosion occuring at the air/sample interface. The stainless steel can be obtained from Torrington Brass and Steel Co., Inc., P.O. Box 986, Torrington, CT 06790; steel part number: T-302, 1/2 hard.

4.4.1.7 Deleterious effects on steel, type III. Descale steel coupons by immersing in concentrated hydrochloric acid. Rinse the coupons, dry, rinse with acetone and keep in a desiccator. Weigh 3 coupons to the nearest milligram and place the coupons in three 250 ml beakers, resting each coupon on a v-shaped piece of 1/8 inch glass tubing. Add 200 ml of cleaner to each beaker and let stand for 24 hours at room temperature. Remove the coupons, rinse with water several times, rinse with distilled water and then with acetone, and dry at 105°C for 15 minutes and weigh to the nearest milligram. Report the percent weight loss as the average of the percent weight losses for each coupon calculated by using the following formula.

$$\text{Percent weight loss} = \frac{B-A}{B} \times 100$$

where. B = weight in milligrams before testing.
A = weight in milligrams after testing.

The steel can be obtained from Torrington Brass and Steel Co., Inc., P.O. Box 986, Torrington, CT 06790; steel part # 1010 steel, # 3 Temper, 0.62" x 1.375".

4.4.1.8 Deleterious effects, porcelain enamel. Pour enough cleaner on a sample of porcelain to cover a 2 inch diameter area. Cover the porcelain with a watch glass and let stand for 18 hours. Examine for evidence of etching. The porcelain can be obtained as Coors # 56003 from the Arthur H. Thomas Co., P.O. Box 779, Philadelphia, PA 19105.

4.4.1.9 Foam stability. A stock solution is made by making up 10 grams of the sample, taken from a depressurized container, to 1 liter with distilled water. Ten ml of this solution shall be placed in a 25 ml graduated cylinder with a stopper. The stoppered cylinder is shaken vigorously for 1 minute. The resulting foam height is reported as the numerical difference between the readings in milliliters at the top of the liquid and at the top of the combined foam and liquid. The foam height is determined at 0 minutes after the shaking and the foam height loss is reported as the difference between the foam heights at 0 minutes and at 5 minutes.

4.4.1.10 Particle size. Weigh 10.00 grams of type II cleaner in a beaker. Add 200 ml of distilled water and digest for 1 hour on a steam bath to dissolve the soluble matter. Weigh a No. 80 sieve to the nearest 0.01 gram. Pour the sample solution through the sieve and into a clean beaker. Pour 200 ml of hot distilled water through the sieve in small portions, while agitating the sieve. Dry the sieve in a forced air oven at 105° to 110°C for 1 hour, cool in a desiccator, and weigh. Transfer the material which passed through the No. 80 sieve to a No. 325 sieve, while agitating the sieve with 200 ml of hot distilled water in small portions followed by 100 ml of acetone in small portions. Dry the sieve in a forced air oven at 105° to 110°C for 1 hour, cool in a desiccator, and weigh.

4.4.1.11 Rinsability. Place 30 ml of the type II cleaner in a clean 500 ml round bottom flask and move the flask about so that the entire spherical interior surface of the flask is covered. Add 50 ml of distilled water, stopper, shake vigorously for 10 seconds, invert the flask and drain for 10 seconds. Repeat with two more 50 ml portions of distilled water. Allow the flask to dry and examine for traces of foreign material adhering to the interior of the flask.

4.4.1.12 Viscosity. Measure the viscosity of the product at 22° + 1°C, using a Brookfield model RV viscometer with a No. 3 spindle at 20 RPM. Convert the readings to centipoises.

4.4.1.13 Non-volatile matter content, type II. Weigh a one gram sample (to + .005 gram) into a tared crucible and dry to constant weight in a forced air oven at 105°C. Constant weight is attained when the weight loss is less than

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:19Z
Check the source to verify that this is the current version before use.

0.1 percent per hour. Calculate the percent non-volatile matter by using the following formula:

$$\text{Percent N.V.M.} = \frac{\text{Weight of residue}}{\text{Weight of sample}} \times 100$$

4.4.1.14 Hydrogen chloride content, type III. Weigh a glass stoppered erlenmeyer flask containing about 30 ml of distilled water. Add 3 to 4 ml of the sample, stopper and reweigh. Dilute to 50 ml with distilled water and titrate with 1 N sodium hydroxide using methyl orange or electrometric indicator. Calculate the percentage of HCl by employing the formula.

$$\text{Percent} = \frac{A \times N \times 0.03647 \times 100}{B}$$

where A = milliliters of sodium hydroxide solution.
N = normality of sodium hydroxide solution.
B = weight in grams of the specimen.

4.4.2 Aerosol testing.

4.4.2.1 Sampling plan. The sample unit shall be defined as one filled Aerosol package. The samples shall be selected in accordance with inspection level S 2 and the AQL shall be 4.0 percent defective.

4.4.2.2 Net weight. Weigh filled container to the nearest gram. Shake the container with a wrist twisting motion for 15 seconds at the approximate rate of one complete cycle per second. Holding the container in the position recommended on the product label, exhaust the container by holding the valve wide open. Rinse the exterior of the container and dry, using a towel and compressed air for drying inaccessible areas including the interior of the actuator. Weigh the exhausted container to the nearest gram. The measured net weight in ounces shall be calculated using the following formula:

$$N.W. = (G - E - 10)/(28.375)$$

where. G = gross weight of filled container (grams).
E = weight of exhausted container (grams).

4.4.2.3 Spray pattern. Activate the valve of a container for a few seconds to stabilize the delivery of the product. Spray for 1 or 2 seconds at a surface 6 inches distant. Examine the distribution of product on the surface and examine the actuator for evidence of excessive deposition or leakage of product.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C, as specified (see 6.2).

5.1.1 Levels A and B.

5.1.1.1 Type I. Seventeen ounces of cleaner and propellant shall be packaged in a pressurized can conforming to PPP-C-96 type IX, class 2. The valve stem shall be fitted with a side discharge plastic actuator shaped and marked to indicate the direction of discharge. When tested as specified in 4.4.2.2, the measured net weight to the nearest ounce shall be no less than 17 ounces. When tested as specified in 4.4.2.3, the spray pattern shall be an evenly distributed disc shape 4 to 8 inches in diameter, and there shall not be evidence of excessive leakage around the actuator.

5.1.1.2 Types II and III. One (1) quart of type II or type III product shall be packaged in a 1 quart plastic bottle commonly used for the item. The bottle shall be closed to prevent leakage. The closure shall have a recloseable feature, such as a folding nozzle or a screw cap closure.

5.1.2 Level C. Cleaning compounds of types I, II, or III shall be packaged to afford adequate protection against physical damage or deterioration during shipment from the supply source to the first receiving agency. The package and quantity per package shall be the same as that normally used by the supplier for retail distribution.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:19Z
Check the source to verify that this is the current version before use.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Twelve (12) cans of type I product shall be packed in a box conforming to PPP-B-636, class weather resistant. The box shall be closed, waterproofed, and banded in accordance with the appendix to the box specification.

5.2.1.2 Types II and III. Twelve (12) bottles of type II or type III product shall be packed in a box conforming to PPP-B-636, class weather resistant. The box shall be provided with dividers, partitions, or separators which are the full height of the interior of the box. The box shall be 3/8 x 1/4 -0 inch higher than the bottles. The box shall be closed, waterproofed, and banded in accordance with the appendix to the box specification.

5.2.2 Level B. The cleaning compound shall be packed as specified in 5.2.1 except that the box shall be class domestic and does not require waterproofing or banding.

5.2.3 Level C. Cleaning compound of one description and quantity only, as specified (see 6.2) shall be packed in a container complying with the Uniform Freight or National Motor Freight Classification, as applicable. The containers shall be packed to assure carrier acceptance and safe arrival at destination.

5.3 Marking. The unit containers (cans and bottles) and shipping containers (boxes) shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129, as applicable, and with 3.11.

6. NOTES

6.1 Intended use.

6.1.1 Type I. The type I product is intended for cleaning organic soils from porcelain and ceramic tile surfaces. The product is safe to use on stainless steel, chrome plated brass and other metals.

6.1.2 Type II. The type II product is intended for cleaning organic and inorganic soils from porcelain and ceramic tile surfaces. The product is safe to use on stainless steel, chrome plated brass and other metals.

6.1.3 Type III. The type III product is intended for removing difficult organic and inorganic soils from porcelain surfaces. This product is not intended for use on porcelain enamel fixtures and must not be allowed to come into contact with metal because of its corrosive nature.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type required.
(c) Level of packing and packaging required (see 5.1 and 5.2).
(d) Applicable marking document (see 5.3).
(e) Whether the product is required to be a sanitizer or disinfectant (see 3.10).

6.3 Supersession. Type III cleaner described in this specification supersedes the product described in Federal Specification P-C-450B.

Military Custodian :

Army - GL

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
VA - DMS
HEW - NIH
GSA - PBO
EPA

PREPARING ACTIVITY: GSA - FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 each.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:19Z
Check the source to verify that this is the current version before use.